MILITARY SPECIFICATION SHEET

ELECTRON TUBES, KLYSTRON

TYPES 8849 and 8821 1/
a b 2/

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Long life, low distortion, extremely stable reflex klystron, designed for reliable service in microwave relay systems.

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Irs	Er	Body T	Ehk
Unit:	V	Vdc	mAdc	Vdc	°C	Vdc
Maximum:	7.0	775	80	-1,000	150	±45
Minimum:	5.7	---	---	-20 (see note 2)	---	---

PHYSICAL CHARACTERISTICS:

Cooling:	Conduction cooled
Cathode:	Oxide coated, unipotential
Dimensions:	See figure 1

TEST CONDITIONS: See note 3

Parameter:	Ef	Ers	Er	F	Body T
Unit:	V	Vdc	Vdc	GHz	°C
a - - -	6.3	750	-200 to -450	7.587	100 (max)
b - - -	6.3	750	-200 to -450	7.275	100 (max)

GENERAL:

Qualification - Required

Holding period (t = 48 hours)

1/ Formerly VA-222A and VA-222B
2/ To identify those tests that are applicable to a given type or to several types; tube types are designated by letters.

8849 8821
a b

FSC 5960

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
1256	Electrode current (beam)	a, b	Er/max Po	4.0	II	Irs	---	80	mAdc
1231	Emission	a, b	Ef = 5.7 V; Er = -350 to -450 Vdc (see notes 1, 4, and 5)	4.0	II	$\frac{\Delta Irs}{Irs}$	---	-15	%
4213	Reflector voltage	b	Er/max Po; F = 7.125 and 7.425 GHz	4.0	II	Er	-250	-400	Vdc
		a	Er/max Po; F = 7.425 and 7.750 GHz	4.0	II	Er	-250	-400	Vdc
4229	Total reflector current	a, b	Er = -350 to -450 Vdc (see notes 1, 5, and 6)	4.0	II	Ir	---	5	μAdc
4223	Mechanical tuning range	b	Er/max Po (see note 7)	4.0	II	F	7.125	7.425	GHz
		a	Er/max Po (see note 7)	4.0	II	F	7.425	7.750	GHz
1301	Heater current	a, b	Ef = 6.3 V (see note 1)	4.0	II	If	---	0.9	A
4250	Power output	b	Er/max Po; F = 7.125 and 7.425 GHz	4.0	II	Po	1.0	---	W
		a	Er/max Po; F = 7.425 and 7.750 GHz	4.0	II	Po	0.7	---	W
4280	Electronic tuning range	b	Er/50% max Po; F = 7.125 and 7.425 GHz	4.0	II	ΔF	28	---	MHz
		a	Er/50% max Po; F = 7.425 and 7.750 GHz	4.0	II	ΔF	25	---	MHz
4231	Hysteresis	a, b	Er/max Po (see note 8)	4.0	II	---	---	50	%
---	Tuning torque	b	F = 7.125 and 7.425 GHz	4.0	II	---	---	32	oz-in.
		a	F = 7.425 and 7.750 GHz	4.0	II	---	---	32	oz-in.

METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 2								
---	Heater frequency modulation	b	Er/max Po; Ef = 6.3 V at 400 Hz; F = 7.125 GHz	---	---	ΔF	---	30	kHz (p-p)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
---	Heater frequency modulation	a	Er/max Po; Ef = 6.3 V at 400 Hz; F = 7.425 GHz	---	---	ΔF	---	30	kHz (p-p)
4027	Temperature coefficient	b	Er/max Po; TA = +25° to +80°C (see note 9); F = 7.125 GHz	---	---	ΔF/ΔTA	---	±100	kHz/°C
4		a	Er/max Po; TA = +25° to +80°C (see note 9); F = 7.425 GHz	---	---	ΔF/ΔTA	---	±100	kHz/°C
---	Reflector voltage modulation coefficient	a, b	Er/max Po; ΔF = 5 MHz (p-p)	---	---	ΔF/ΔEr	225	525	kHz/V
---	Beam voltage modulation coefficient	a, b	Er/max Po; ΔErs = +9 Vdc	---	---	ΔF/ΔErs	---	±150	kHz/V
	Quality conformance inspection, part 3								
---	Life-test provisions	b	Group D; power output, F = 7.275 GHz	---	---	t	1,000	---	hrs
---	Life-test end point	b	Er/max Po	---	---	Po	0.8	---	W
---	Life-test provisions	a	Group D; power output, F = 7.587 GHz	---	---	t	1,000	---	hrs
---	Life-test end point	a	Er/max Po	---	---	Po	0.8	---	W

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. The reflector shall always be at least 20 volts negative with respect to the cathode.
3. All oscillation tests shall be made with the tube rigidly connected to a CMR-137 flange on RG-50/U waveguide equipment. The load VSWR for the tube shall be less than 1.1:1. The temperature shall be measured at the junction of the cathode shell and the heat sink as shown on figure 1. For maximum tube life, the temperature should be maintained below 100° C.
4. The beam current shall not drop more than the specified percentage when the heater voltage is decreased from the normal operating potential to the lower specified value.
5. The tube shall not be oscillating during this test.
6. After 2 minutes with all voltages applied, total reflector current shall not exceed the specified limit.
7. At the completion of final tests, set frequency to the specified lower limit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.

NOTES: - Continued

8. A power meter shall be used to measure the relative power levels instead of the specified calibrated crystal.

9. The temperature coefficient, when tested under the conditions specified, shall be computed in kilohertz per degree centigrade over the temperature range specified.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL
Navy -
Air Force - 11, 17, 85
DSA - ES

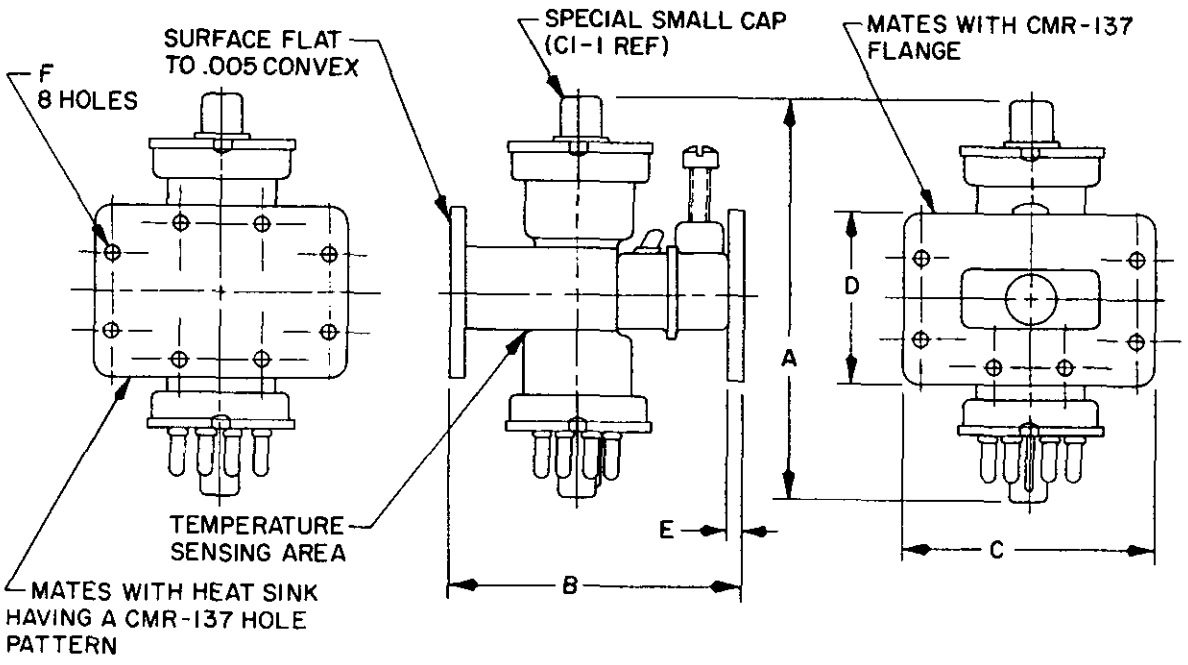
User activities:
Army - WC, ME
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2396)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.



PIN CONNECTIONS

1	sh
3	nc
4	k
5	h
6	h
7	no con
cap	r

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A		3.750 (95.25)
B	2.450 (62.23)	2.485 (63.12)
C		2.295 (58.29)
Quality conformance inspection, part 2		
D		1.545 (39.24)
E	.105 (2.67)	.135 (3.43)
F	No. 6-32 UNC-2A	

FIGURE 1. Outline drawing of electron tube types VA-222A and VA-222B.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:15Z
Check the source to verify that this is the current version before use.